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Weighted Decision Matrices Guide

A. Affordability

B. Simplicity of making turns

C. Weight

D. Ease of construction and assembly

E. Uniqueness

	Criterion A	Criterion B	Criterion C	Criterion D	Criterion E	Row Total
Criterion A	--	1	1	0	1	3
Criterion B	0	--	1	1	0	2
Criterion C	0	0	--	0	1	1
Criterion D	1	0	1	--	1	3
Criterion E	0	1	0	0	--	1

After discussing each comparison as a team, a completed table could look like the following:

	Criterion A	Criterion B	Criterion C	Criterion D	Criterion E	Row Total
Criterion A	--	1	1	0	1	3
Criterion B	0	--	1	1	0	2
Criterion C	0	0	--	0	1	1
Criterion D	1	0	1	--	1	3
Criterion E	0	1	0	0	--	1

The result is a relative ranking of criteria. The scores should not be used to directly determine percentages, as they do not take into account the difference between criteria

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with the same scores. The team can differentiate between ties by considering which one was considered more important in the head-on comparison in the table. For example, Criteria A and D are ranked highest overall with scores of 3, but when compared directly to each other in the table, Criterion D was deemed more important.

The team can then use the relative rankings to better inform the relative assignment of weights to each criterion. For example, A and D are scored the same but D should receive a slightly higher weight than D because of the tie-breaker. A's score of 3 beats B's score of 2, and the difference in weights should be more than the difference in weights between D and A. A reasonable comparison scale could be as follows:

	E	C			B			A	D
1	2	3	4	5	6	7	8	9	10

This scale does not have a zero value for anchoring the values, but it could nonetheless form a rough basis for determining weights. In one such estimation, the team could add the five new scores (2+3+6+9+10=30) and take each score as a percentage, resulting in the following:

D:33%, A:30%, B:20%, C:10%, E:7%

The result of this entire process is not a formula for determining the weights, but a reasonable guide to aid in the assignment of weights.

Setting Up and Using a Weighted Decision Matrix

Now that the criteria have weights, Team Turtle can set up the weighted decision matrix below:

	weight	Option 1	Option 2	Option 3	Option 4
Criterion D	0.33	2	-3	4	1
Criterion A	0.30	0	1	2	3
Criterion B	0.20	3	-2	0	-1
Criterion C	0.10	-1	4	3	-2
Criterion E	0.07	1	1	-5	0
score		1.23	-.62	1.87	.83

engineering 4 us all

Student Handout

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Each design option receives a score for each criterion. The score is multiplied by the corresponding criterion's weight, and these weighted values are added for a total design score.

This table should not be filled out by group consensus. This could leave less vocal members without input in this stage of the decision-making process. Instead, each member should be permitted to fill out this table on his or her own. Each cell in the team's master table is populated by adding together the scores from each member's corresponding cell, essentially creating an average of the members' scores.

The team can choose any scale it wishes for each member to assign scores. A few examples include:

- Assigning a score on a scale of 1-5. Design options can receive the same score.
- Assigning a rank on a scale of 1 to the # of design options being considered. Each design option must be ranked differently.
- Assigning a -1, 0, or +1 to indicate a design option is poor, neutral, or good according to the given criteria.

For example, each team member could be asked to fill out the table using the -1, 0, +1 scale. The sum of scores for all eight members might result in the following master table:

	weight	Option 1	Option 2	Option 3	Option 4
Criterion D	0.33	2	-3	4	1
Criterion A	0.30	0	1	2	3
Criterion B	0.20	3	-2	0	-1
Criterion C	0.10	-1	4	3	-2
Criterion E	0.07	1	1	-5	0
score		1.23	-0.62	1.87	0.83

Option 3 would be the winning choice for Team Turtle.